Linear Algebra

Unit 6: Eigenvalues and Eigenvectors

Practice Test — Study Copy

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 6 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Eigenvector _______
2. Eigenvalue _______
3. Characteristic equation _______
4. Diagonalization _______
5. Diagonalizable matrix _______
6. Symmetric matrix _______
7. Positive definite matrix _______
8. Spectral theorem _______

(A) The scalar λ in $A\mathbf{x} = \lambda\mathbf{x}$: the factor by which its eigenvector is stretched.

(B) A nonzero vector $\mathbf{x}$ whose direction is unchanged by A — $A\mathbf{x}$ is just a multiple of $\mathbf{x}$.

(C) The equation $\det(A - \lambda I) = 0$, whose solutions are the eigenvalues.

(D) Writing $A = X\Lambda X^{-1}$ with the eigenvectors in X and the eigenvalues down Λ.

(E) A square matrix with enough (n) independent eigenvectors to fill the columns of X.

(F) A matrix equal to its own transpose, $S = S^{\mathsf{T}}$.

(G) A symmetric matrix whose eigenvalues are all positive (its energy $\mathbf{x}^{\mathsf{T}}S\mathbf{x} > 0$).

(H) The guarantee that a symmetric matrix has real eigenvalues and perpendicular eigenvectors.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. If $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{x}$ is an (A) eigenvector (B) pivot (C) determinant (D) inverse
2. The eigenvalues of A are the solutions of (A) $A\mathbf{x} = \mathbf{0}$ (B) $\det A = 0$ (C) $\det(A - \lambda I) = 0$ (D) $A = A^{\mathsf{T}}$
3. The eigenvalues of a triangular matrix are (A) all 1 (B) its diagonal entries (C) its column sums (D) always 0
4. A square matrix is diagonalizable exactly when it has (A) determinant 1 (B) a zero eigenvalue (C) equal rows (D) n independent eigenvectors
5. A symmetric matrix $S = S^{\mathsf{T}}$ always has (A) complex eigenvalues (B) perpendicular eigenvectors (C) determinant 0 (D) no eigenvectors
6. The 2×2 symmetric matrix $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite when (A) $a < 0$ (B) $ac - b^2 = 0$ (C) $a > 0$ and $ac - b^2 > 0$ (D) $a = c$

Part C — Short Answer & Computation (35 pts)

Show your work.

1. Let $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$. **(a)** Is $\mathbf{x} = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ an eigenvector? If so, give its eigenvalue. **(b)** Is $\mathbf{y} = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$ an eigenvector? Explain how you can tell.

2. Find the eigenvalues of $B = \left[\begin{smallmatrix} 4 & 2 \\ 1 & 3 \end{smallmatrix}\right]$ by writing and solving its **characteristic equation** $\det(B - \lambda I) = 0$.

3. For that same $B = \left[\begin{smallmatrix} 4 & 2 \\ 1 & 3 \end{smallmatrix}\right]$, find an **eigenvector** for each eigenvalue. Then check your eigenvalues with the **trace** and **determinant** of B.

4. Let $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, whose eigenvalues are $\lambda = 3, 1$ with eigenvectors $\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$. Use $A^k = X\Lambda^k X^{-1}$ to compute A^2.

5. Decide whether each matrix is **diagonalizable**, and explain why: **(a)** $P = \left[\begin{smallmatrix} 3 & 0 \\ 1 & 3 \end{smallmatrix}\right]$; **(b)** $Q = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.

6. Let $S = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$. **(a)** Show S is **positive definite** using the test $a > 0$ and $ac - b^2 > 0$. **(b)** Without computing them, explain why the eigenvectors of S are perpendicular.

7. A system $\dfrac{d\mathbf{u}}{dt} = A\mathbf{u}$ has eigenvalues $\lambda_1 = 3$, $\lambda_2 = -1$ with eigenvectors $\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$, and starting vector $\mathbf{u}(0) = \left[\begin{smallmatrix} 5 \\ 1 \end{smallmatrix}\right]$. **(a)** Write $\mathbf{u}(0) = c_1\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right] + c_2\left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$. **(b)** Write the general solution $\mathbf{u}(t)$. **(c)** Does $\mathbf{u}(t) \to \mathbf{0}$ (is it

stable)? Explain.

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. To find eigenvalues we solve $\det(A - \lambda I) = 0$. **(a)** Starting from $A\mathbf{x} = \lambda\mathbf{x}$, explain why an eigenvalue makes the matrix $A - \lambda I$ *singular*. **(b)** Explain why "singular" is exactly the same as "$\det(A - \lambda I) = 0$", so that solving that equation finds the eigenvalues.

2. **(a)** Once $A = X\Lambda X^{-1}$, explain why $A^k = X\Lambda^k X^{-1}$ — and why that makes high powers of A easy to compute. **(b)** A matrix S is symmetric. Explain why its eigenvectors are automatically perpendicular, and why "all eigenvalues positive" guarantees S is invertible.